

Eisenhower Task Manager - Project Overview Document (POD)

Last Updated: January 27, 2026 **Current Version:** 2.0.1 **Status:** ☒ Production Ready **License:** MIT

Executive Summary

The **Eisenhower Task Manager** is a fully-featured productivity application implementing the Eisenhower Decision Matrix (Urgent/Important framework). It provides an intuitive interface for task planning, tracking, and analysis with powerful visualization and scoring capabilities.

Current State: Fully functional, feature-complete v2 application running on React 18 + Vite with local file-based or localStorage persistence.

Key Metrics

Metric	Value
Current Version	2.0.1
Status	Production Ready
Lines of Code (App)	2,284 (src/App.jsx)
Lines of Code (PIN)	114 (src/components/PINModal.jsx)
Backend (Server)	117 lines (Express.js)
Build Size	~202 KB (gzipped: ~60 KB)
Deployment	GitHub Pages + Local Development
Tech Stack	React 18, Vite, Tailwind CSS, Lucide Icons

Features Implemented

Core Task Management

- ☒ **Task Creation & Editing** - Full CRUD operations with rich metadata
- ☒ **Eisenhower Matrix View** - 4-quadrant prioritization (Urgent/Important)
- ☒ **List View** - Sortable, filterable table interface
- ☒ **Gantt Chart View** - Timeline visualization with time estimates
- ☒ **Task Recurrence** - Once, Daily, Weekly, Monthly, Yearly patterns
- ☒ **Task Categories & Subcategories** - Customizable organization
- ☒ **Progress Tracking** - Percentage completion for long-running tasks

Analytics & Scoring

- ☒ **Planning/Execution Score** - Measures how accurately you estimate due dates
 - Formula: $(\text{Due Date} - \text{Completed Date}) / (\text{Due Date} - \text{Assigned Date})$
 - Visual representation with color-coded bars (green = early, red = late)
- ☒ **Quality Rating** - 1-5 star quality assessment on completion
- ☒ **Ease Rating** - 1-5 star difficulty assessment on completion
- ☒ **Time Estimates** - Hours or days estimation for Gantt visualization

User Interface

- ☒ **Beautiful, Modern Design** - Tailwind CSS with gradient effects
- ☒ **Responsive Layout** - Works on desktop, tablet, mobile
- ☒ **Multiple Views** - Matrix, List, Gantt with smooth transitions
- ☒ **PIN Protection** - Session-based courtesy lock (default PIN: 1234)
- ☒ **Real-time Updates** - Instant UI refresh on task changes
- ☒ **Color-Coded Quadrants:**
 -  Red: Do First (Urgent + Important)
 -  Blue: Schedule (Not Urgent + Important)
 -  Amber: Delegate (Urgent + Not Important)
 -  Gray: Eliminate (Not Urgent + Not Important)

Data Management

- ☒ **localStorage Support** - Browser-based persistence (simple mode)
- ☒ **File-Based Storage** - JSON files in `data/` folder (production mode)
- ☒ **Auto-Save** - Never lose your work with automatic persistence
- ☒ **Export/Import** - JSON backup and restore functionality
- ☒ **Backup Metadata** - Track export history and last save timestamps
- ☒ **Data Validation** - Zod schema validation on data structures

Advanced Filtering & Organization

- ☒ **Filter by Quadrant** - All, Do First, Schedule, Delegate, Eliminate
- ☒ **Filter by Category** - Custom categories (Career, Personal, Health, etc.)
- ☒ **Filter by Status** - Active, Completed, All tasks
- ☒ **Filter by Recurrence** - Once, Daily, Weekly, Monthly, Yearly, All
- ☒ **Sort Options** - Priority, Due Date, Category, Recurrence
- ☒ **Search Capability** - (Implemented in List view)



Architecture

Technology Stack

```
Frontend:
├─ React 18.2.0           # UI Framework
├─ Vite 4.3.9             # Build tool & dev server
├─ Lucide React 0.263.1   # Icon library
├─ Tailwind CSS 3.3.0     # Styling (core + utilities)
└─ PostCSS 8.4.24        # CSS processing

Backend:
├─ Express 4.18.2        # HTTP server
├─ CORS 2.8.5            # Cross-origin middleware
└─ fs-extra 11.1.1       # File system utilities

Development:
├─ Concurrently 8.2.0    # Parallel process runner
└─ (No build-time dependencies)
```

Project Structure

```
eisenhower-task-manager-v2/
├─ eisenhower-app/
│  └─ src/
│     └─ App.jsx           # Main application (2,284 lines)
│     └─ main.jsx          # React entry point
│     └─ index.css         # Base styles
│     └─ components/
│        └─ PINModal.jsx   # PIN lock component (114 lines)
│  └─ data/                # Task data files (production mode)
│     └─ tasks.json
│     └─ settings.json
│     └─ backup-metadata.json
│  └─ dist/                # Production build output
│  └─ server.js            # Express backend (117 lines)
│  └─ package.json
│  └─ vite.config.js
│  └─ tailwind.config.js
│  └─ postcss.config.js
│  └─ index.html
│  └─ start.bat            # Windows quick-start script
│  └─ start.sh             # Mac/Linux quick-start script
└─ [documentation files]
   └─ README.md
   └─ CHANGELOG_V2.md
   └─ GANTT_IMPLEMENTATION_SUMMARY.md
   └─ SCORE_FEATURE_GUIDE.md
   └─ [other guides]
```

Data Model

Task Structure

```
{
  id: string (UUID),
  task: string,
  category: string,
  subcategory: string,
  isUrgent: boolean,
  isNecessary: boolean,
  rank: number (1-3 within quadrant),
  assignedDate: string (YYYY-MM-DD),
  dueDate: string (YYYY-MM-DD),
  completedDate: string | null (YYYY-MM-DD),
  percentComplete: number (0-100),
  isRecurring: boolean,
  recurringPattern: string ('once' | 'daily' | 'weekly' | 'monthly' | 'yearly'),
  notes: string,
  qualityRating: number | null (1-5),
  easeRating: number | null (1-5),
  timeEstimateValue: number | null,
  timeEstimateUnit: string ('hours' | 'days')
}
```

Settings Structure

```
{
  categories: string[],           // e.g., ['Career', 'Personal', 'Health']
  subcategories: {                // Organized by category
    [category]: string[]
  }
}
```

Backup Metadata Structure

```
{
  lastExport: string | null,      // ISO timestamp
  exportCount: number,           // Total exports
  lastAutoSave: string | null    // ISO timestamp
}
```

Deployment & Availability

Current Deployment

- **Live URL:** <https://www.peirastes.com/projects/eisenhower-task-manager.html>
- **Deployment Method:** GitHub Pages (dist/ folder)

- **Update Frequency:** Manual (rebuild and push dist/)
- **Availability:** 24/7 with GitHub Pages SLA

Local Development

- **Dev Server:** `npm run dev` → http://localhost:5173
- **Production Build:** `npm run build` → creates dist/
- **File-Based Server:** `npm start` → runs Express on port 3001



Recent Changes (v2.0 → v2.0.1)

v2.0.1 (January 2026)

- ☒ Planning/Execution Score feature finalized
- ☒ Gantt chart with time estimations
- ☒ Quality & Ease rating system
- ☒ Completion verification modal
- ☒ Enhanced filtering and sorting
- ☒ All views fully functional and tested

v2.0 (Initial Release)

- Core Eisenhower Matrix implementation
- Multiple view modes (Matrix, List, Gantt)
- PIN protection
- Export/import functionality
- Recurrence patterns



How to Use

Quick Start (Recommended)

Windows:

```
cd eisenhower-task-manager-v2/eisenhower-app
start.bat
```

Mac/Linux:

```
cd eisenhower-task-manager-v2/eisenhower-app
chmod +x start.sh
./start.sh
```

Manual Setup

```
# Install dependencies
npm install

# Run in simple mode (localStorage)
npm run dev

# OR run in file-based mode (data/ folder)
npm start
```

Storage Modes

Mode	Command	Storage	Best For
Simple	npm run dev	localStorage	Quick testing, single device
File-Based	npm start	data/ folder	Production, cloud sync, backups

Data Sync with Cloud Storage

To enable automatic sync across devices:

1. Place the entire project folder in Dropbox/Google Drive/iCloud
2. All data files (tasks.json, settings.json) automatically sync
3. Works seamlessly on multiple machines

 Security Notes

PIN Protection

- **Level:** Courtesy protection only
- **Default PIN:** 1234
- **Method:** Frontend session-based (sessionStorage)
- **Bypass:** Possible via browser DevTools

- **Use Case:** Casual/accidental access prevention, not cryptographic security

Data Privacy

- ☒ All processing done locally (no cloud, no analytics)
- ☒ Data stored in plaintext JSON (self-hosted)
- ☒ No telemetry or external API calls
- ☒ Full data ownership
- ☒ Not encrypted (can be encrypted by user if needed)

Recommendations

- ⚠ Back up data regularly (use Export feature)
- ⚠ Don't store sensitive information (password, credit card, etc.)
- ⚠ Use OS-level encryption for device security
- ⚠ Keep browser and Node.js updated

Development

Available Commands

```
# Development
npm run dev          # Start Vite dev server (http://localhost:5173)
npm start            # Start Express backend + Vite (concurrent)

# Production
npm run build         # Build for production (creates dist/)
npm run preview       # Preview production build locally

# Backend Only
npm run server        # Start Express server only (port 3001)
```

File Structure for Development

```
Key Files to Edit:
├─ src/App.jsx          # Main component (all features)
├─ src/components/      # Reusable components
│   └─ PINModal.jsx     # PIN screen
├─ server.js            # API endpoints
├─ tailwind.config.js   # Color/style config
└─ data/settings.json   # Default categories
```


Common Customizations

Change PIN: 1. Edit `src/components/PINModal.jsx` 2. Change `CORRECT_PIN` constant 3. Rebuild: `npm run build`

Change Default Categories: 1. Edit `data/settings.json` 2. Or modify default in App.jsx state initialization 3. Restart app

Change Colors: 1. Edit `tailwind.config.js` 2. Modify quadrant color classes (bg-red-500, etc.) 3. Rebuild

Metrics & Performance

Build Performance

Build Time:	~6.5 seconds (Vite)
Output Size:	202 KB (JavaScript)
Gzip Size:	~60 KB (actual transfer)
CSS Size:	27 KB

Runtime Performance

- ✔ Instant task creation/editing
- ✔ Smooth view transitions
- ✔ No lag with 100+ tasks
- ✔ Fast filtering and sorting
- ✔ Responsive UI on modern hardware

Browser Compatibility

- ✔ Chrome/Edge 90+
- ✔ Firefox 88+
- ✔ Safari 14+
- ✔ Mobile browsers (iOS Safari, Chrome Mobile)

Known Limitations

Current Limitations

- Single User** - No multi-user support (localhost only)

2. **No Cloud Sync** - Manual sync via file sharing or cloud storage folder
3. **No Server Backup** - Data stored locally only
4. **No Mobile App** - Web-only (responsive design)
5. **PIN is Not Secure** - Courtesy protection only, not cryptographic
6. **No Recurrence Automation** - Must manually mark recurring tasks complete

Workarounds

- Use cloud folder (Dropbox, Google Drive) for cross-device sync
- Manually export backups weekly
- Use iOS/Android app wrappers for “app-like” experience if needed

Future Enhancement Opportunities

Short-term (1-2 months)

- ☐ Recurrence automation (auto-create next occurrence)
- ☐ Task templates (save common task patterns)
- ☐ Keyboard shortcuts (vi keybindings, etc.)
- ☐ Dark mode theme
- ☐ Mobile app wrapper (Electron/Capacitor)

Medium-term (2-6 months)

- ☐ Multi-user backend with PostgreSQL
- ☐ Cloud sync with conflict resolution
- ☐ Analytics dashboard (trends, patterns)
- ☐ Collaborative features (sharing, permissions)
- ☐ Mobile app (iOS/Android native)
- ☐ Calendar integration (Google Calendar, Outlook)

Long-term (6+ months)

- ☐ Payment system for commercial version
- ☐ Team collaboration features
- ☐ AI-powered suggestions
- ☐ Advanced analytics and reporting
- ☐ Integration marketplace (Slack, Teams, etc.)

Documentation

Included Documentation

Document	Purpose	Audience
README.md	User guide and quick start	End users
GANTT_IMPLEMENTATION_SUMMARY.md	Technical deep-dive on Gantt chart	Developers
SCORE_FEATURE_GUIDE.md	Explanation of Planning/Execution Score	Users & analysts
CHANGELOG_V2.md	Version history and changes	All
UPDATES_MODIFICATIONS_LIST.md	Detailed change log	Developers
SCORE_FEATURE_IDEAS.md	Future score-related features	Product team

External Resources

- **Eisenhower Method:** https://en.wikipedia.org/wiki/Time_management#Eisenhower_matrix
- **React Docs:** <https://react.dev/>
- **Vite Docs:** <https://vitejs.dev/>
- **Tailwind CSS:** <https://tailwindcss.com/>

Support & Maintenance

Getting Help

1. **Check README.md** - Most common issues covered
2. **Review documentation files** - Feature guides
3. **Browser console** - Look for error messages (F12)
4. **Check GitHub Issues** - (if shared publicly)

Maintenance Tasks

Weekly: - No automated tasks required - Optional: Manual data backup

Monthly: - Review task backlog - Check for browser updates - Update Node.js if major version released

Learning & Development

Code Quality

-  Modular React component architecture
-  Clean state management patterns
-  Reusable utility functions
-  Clear variable naming
-  Comments on complex logic

Code Organization

```
// Main App component structure:  
- PIN verification  
- State management (tasks, settings, view, filters)  
- Helper functions (calculateScore, getQuadrant, etc.)  
- Sub-components (inline for simplicity)  
  - EisenhowerTaskManager (main wrapper)  
  - MatrixView (4-quadrant layout)  
  - ListView (table view)  
  - GanttView (timeline)  
  - TaskCard (individual task)  
  - TaskForm (create/edit modal)  
  - CompletionModal (rating prompt)
```

Areas for Learning

- **React Patterns:** State management, conditional rendering, list handling
- **Vite:** Modern build tooling, dev server, HMR
- **Tailwind CSS:** Utility-first CSS, responsive design
- **Local Storage:** Browser APIs for persistence
- **Date Handling:** JavaScript Date object, ISO formatting

Verification Checklist

Development Environment: -  Node.js 16+ installed -  npm installed and functional -  npm dependencies installed (`node_modules/`) -  Dev server runs without errors (`npm run dev`)

Application Functionality: - ☒ PIN screen appears and accepts 1234 - ☒ Can create new tasks - ☒ Tasks appear in Matrix view - ☒ Can switch between Matrix/List/Gantt views - ☒ Filtering works correctly - ☒ Completion ratings modal appears - ☒ Export generates JSON file - ☒ Import loads JSON file correctly

Data Persistence: - ☒ localStorage mode: Tasks persist in browser storage - ☒ File-based mode: Tasks saved to data/ folder - ☒ Auto-save works (no manual save button needed) - ☒ Browser refresh doesn't lose data

Production Deployment: - ☒ Build completes without errors (`npm run build`) - ☒ dist/ folder generated - ☒ GitHub Pages deployment successful - ☒ Live site accessible at peirastes.com URL

Conclusion

The **Eisenhower Task Manager v2.0.1** is a fully-featured, production-ready productivity application. It successfully implements the Eisenhower Decision Matrix with powerful analytics, multiple visualization modes, and flexible data management.

Status: ☒ **Complete and Stable Quality:** ☒ **Production Ready Maintenance:** ☒ **Actively Maintained User Experience:** ☒ **Polished and Intuitive**

The application is ready for: - ☒ Personal daily use - ☒ Team productivity workshops - ☒ Educational demonstrations - ☒ Commercial deployment (with enhancements) - ☒ Open-source sharing

Last Updated: January 27, 2026 **Version:** 2.0.1 **Next Review:** Q2 2026